

# Committee Feedback Response

## April 2026

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Random Effects, Post-Exit Dynamics, Never-Exposed DiD, and Propagation Pathways

Ahwaz Akhtar · George Washington University · May 2026

# What the Committee Asked

Five concerns across three domains, all addressed in this update except humidity

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## Econometric

- **Check random effects** as an alternative to FE
- **Lag / exit tests**: what happens when a county leaves a shock state?
- Find **natural experiments** for DiD — are there never-exposed counties?

## General

- **Propagation pathways** must be backed by evidence (heat → delayed care; cold → utilization shifts)

## Environmental

- Add **humidity** (PRISM tdmean) at state level

## Response structure

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| Concern              | Status        |
|----------------------|---------------|
| Random effects       | Done          |
| Post-exit dynamics   | Done          |
| Never-exposed DiD    | Done          |
| Propagation pathways | Done          |
| Humidity (PRISM)     | <b>Parked</b> |

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# Random-Effects Robustness

Hausman tests across all primary state and county outcomes

# Random Effects Is Rejected; Headline Findings Survive

Hausman test on primary FE specs; State + Year two-way effects matched in plm

## Design

- `plm::plm(model="within"|"random",  
effect="twoways")` mirroring the  
`fixest::feols` primary specs
- Hausman `phtest` on each FE/RE pair
- 11 outcomes estimable (4 state, 7 county); 2 state outcomes failed RE fit on thin panels

## Verdict

- **RE rejected for every estimable outcome at  $p < 10^{-9}$**
- Headline state findings (Drought lag-2, Goldilocks 1) have not been rejected in RE

## Hausman test results

| Outcome               | $\chi^2$ | p-value  |
|-----------------------|----------|----------|
| <i>State</i>          |          |          |
| Medical Debt Share    | 233.6    | 4.5e-40  |
| Total PC Health Exp   | 247.9    | 5.7e-43  |
| Medicaid PE Health    | 81.3     | 2.3e-10  |
| Exp                   |          |          |
| Medicare PE Health    | 834.4    | 2.1e-166 |
| Exp                   |          |          |
| <i>County</i>         |          |          |
| Medical Debt Share    | 23905    | <1e-300  |
| Benchmark Silver Real | 23545    | <1e-300  |
| PCPI Real             | 5809     | <1e-300  |
| Civilian Employed     | 220.2    | 4.6e-41  |

Source: `Analysis/random_effects_hausman.csv`

# Post-Exit Dynamics

When a county leaves shock state, what happens to spending?

## Post-Exit Local Projections, $h = 0, 1, 2, 3$

Block A:  $Y_{i,t+h} \sim \text{Exit}_{i,t} \mid \text{fips} + \text{Year}$ ; Block B:  $\text{Shock}_{i,t-1} \times \text{NoShock}_{i,t}$  interaction

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### Drought\_Exit

- **PCPI partial scarring:** +\$1,044 p.c. at  $h = 2$  ( $p = 0.0002$ )
- Recovery rebounds but does not fully restore pre-shock income
- Confirmed by Block B interaction (+964 at  $h = 2$ ,  $p = 0.010$ )

### HDD\_Exit

- **Immediate hospital-cost relief:** -3.13 per-capita at  $h = 0$  ( $p = 0.014$ )
- Sustained at  $h = 2$ : -2.88 ( $p = 0.034$ )
- Cold-shock cost is in the **exposure**, not lasting damage

### CDD\_Exit

- Persistent employment benefits after exit
- Likely part secular regional trend (caveated in synthesis)
- Phase 2 LP rejects within-county heat scarring

Output: Analysis/delta\_coefs.csv (896  $\rightarrow$  1664 rows, +768 Exit-LP cells); 42 plots in Analysis/plots/delta\_exit\_dynamics/

# Diff-in-Diff with Never-Exposed Controls

Sharp natural experiment + Callaway-Sant'Anna long-run profiles

# Phase 0: Are There Never-Exposed Counties?

Inventory of zero-event counties over 2011–2023; feasibility memo selected primary cohorts

## Never-exposed pool by shock

| Shock                             | Total | Never-exposed | Share       |
|-----------------------------------|-------|---------------|-------------|
| Extreme Drought (PDSI $\leq -4$ ) | 3,225 | <b>2,534</b>  | 78.6%       |
| High HDD (top quintile)           | 3,225 | <b>2,303</b>  | 71.4%       |
| High CDD (top quintile)           | 3,225 | <b>2,121</b>  | 65.8%       |
| High AQI Max ( $>100$ )           | 3,225 | 100           | <b>3.1%</b> |

- Drought, HDD, CDD all have large never-exposed pools
- **AQI dropped from DiD**: 3.1% control pool too thin

## Candidate sharp events

| Event                       | Onsets | Onset rate |
|-----------------------------|--------|------------|
| <b>2012 Midwest Drought</b> | 139    | 4.4%       |
| 2022 Drought                | 110    | 3.5%       |
| <b>2013 HDD (cold)</b>      | 407    | 13.0%      |
| 2018 CDD (heat)             | 250    | 8.0%       |
| 2023 AQI (wildfire smoke)   | 336    | 34.1%      |

Source:

Analysis/never\_exposed\_event\_year.csv



# 2012 Midwest Drought: The Cleanest Natural Experiment

139 first-event counties (treatment) vs. 2,534 never-exposed (control), 2011–2023

## Design

$$Y_{it} = \alpha_i + \gamma_t + \tau (\text{Treated}_i \times \text{Post}_t) + \varepsilon_{it}$$

- Treated: first-ever Extreme Drought event = 2012
- Control: never experienced Extreme Drought in panel window
- Cluster: state; SE: `feols(..., cluster="State")`

## Why this is the cleanest piece in the thesis

- Within-county LP cannot rule out recurring-shock confounding
- 2x2 against never-exposed **closes that loophole**
- Drought 2012 is a documented historical event with well-defined geography

## 2x2 ATT results

| Outcome                  | ATT             | p                |
|--------------------------|-----------------|------------------|
| Medical Debt Share       | −0.0062         | 0.49             |
| Hosp Bad Debt PC         | +6.26           | 0.29             |
| <b>PCPI Real</b>         | <b>−\$1,311</b> | <b>0.027</b>     |
| Med HH Income Real       | −\$992          | 0.24             |
| <b>Civilian Employed</b> | <b>−2,053</b>   | <b>0.0001</b>    |
| Benchmark Silver Real    | —               | ACA collinearity |

Source:

`Analysis/did/did_2x2_drought_2012.csv`

# Callaway-Sant'Anna: Long-Run Cold-State Scarring

Manual CS estimator via `fixest` (R `did` package unavailable); never-treated controls; cohort-weighted event-time profile

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## Drought (4 cohorts, 268 treated)

- PCPI at  $e = 0$ :  $-\$1,050$  ( $p = 0.002$ )
- Civilian Employed at  $e = 0$ :  $-142$  ( $p = 0.002$ )
- Civilian Employed at  $e = 11$ :  $-3,915$  ( $p = 0.0001$ )

## HDD (2 cohorts, 230 treated)

- Civilian Employed at  $e = 10$ :  $-4,982$  ( $p = 0.003$ )
- Medical Debt Share at  $e = 10$ :  $+4.9$  **pp** ( $p = 0.0002$ )
- Cumulative scarring through  $e = 6-10$

## CDD: noisy, regional confounds

- Treatment concentrated in TX/GA/MS/FL
- Controls dominated by northern states
- Signs reverse across horizons — **de-emphasize**

## Output

- `Analysis/did/did_cs_att_gt.csv`  
(427 ATT(g,t) cells)
- `did_cs_event_time.csv`  
(cohort-weighted profiles)

# LP vs. DiD: Apparent Tension, Real Complementarity

The two designs answer different questions; the cold-scarring puzzle resolves

# LP and DiD Are Complementary, Not Contradictory

Within-county dynamics vs. persistent treated-vs-never-exposed gap

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## Local Projections / Exit-LP

- Estimand: within-county response when shock state changes year-over-year
- HDD\_Exit → hospital-cost relief at  $h = 0$
- **Read:** no within-county cold scarring after recovery

## CS-DiD with never-exposed

- Estimand: persistent gap between ever-treated and never-treated populations
- HDD long-run:  $-4,982$  jobs and  $+4.9$ pp debt share at  $e = 10$
- **Read:** treated counties end up worse off long-run

**Reconciliation:** cold-shock counties *recover* when shocks end (LP), but they *spend more years in shock state* than never-exposed counties do, so the **population gap accumulates** (CS-DiD).  
Both are correct; they measure different things.

# Propagation Pathways

Evidence-backed mechanisms linking climate shocks to health-finance outcomes

## Pathway-to-Empirics Mapping

Each headline channel is now anchored to 2–4 peer-reviewed references with direction and time-scale

| Pathway                     | Time-scale claim                                         | Where identified                                | Strength         |
|-----------------------------|----------------------------------------------------------|-------------------------------------------------|------------------|
| Heat → delayed care         | Within-year deferral; lag-1/2 utilization                | County Spec 2, CS-DiD CDD                       | Moderate         |
| Cold → utilization shift    | Contemporaneous + lag-1; long-run scarring at $e \geq 8$ | State cold_shock_lag1, CS-DiD HDD               | Strong           |
| Drought → income → debt     | Lag-1, lag-2 income; debt follows                        | State drought_lag2, 2012 DiD, Exit-LP recovery  | <b>Strongest</b> |
| AQI → respiratory / cardiac | Within-year, episodic spikes                             | County continuous AQI, event-study High_AQI_Max | Limited          |

### Selected references

- **Drought → income:** Hornbeck (2012, AER); Burke-Hsiang-Miguel (2015, Nature); Carleton et al. (2022, QJE)
- **Cold:** Deschênes-Moretti (2009, REStat); Gasparrini et al. (2015, Lancet); Andrews et al. (2017)
- **Heat / delayed care:** S. et al. (2021), White (2017, JAEDE), K. et al. (2018, JEFM)

# Humidity: Parked Pending Tooling

PRISM `tdmean` endpoint identified; raster aggregation packages not installed

# Humidity Integration — Status and Resumption Plan

Phase 4 deferred; the highest-priority residual gap from committee feedback

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## What we know

- Endpoint:  
`services.nacse.org/prism/data/get/us/4km/needant/crrayms`, tigris in R
- Returns zipped BIL raster  
(confirmed HTTP 200)
- No state-aggregate API; must  
aggregate locally
- Estimated 408 monthly rasters; ~1.5  
GB on disk

## Blocker

- Raster → state polygon aggregation
- None installed in local R 4.2.2;  
Windows install pulls heavy system  
deps (GDAL/GEOS/PROJ)

## Resumption (when packages available)

- Build  
`download_prism_humidity.R`,  
`process_state_humidity.R`
- Integrate into state pipeline; re-run  
`run_analysis.R`



# What Changed Substantively

New evidence, sharper identification, resolved puzzles

# The Analysis Now Has Two Pieces of Stronger Identification

And one genuine puzzle resolved by treating LP and DiD as complementary

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## New evidence

- **2012 Drought 2x2** is now the cleanest single identification piece (PCPI  $-\$1,311$ ; Employed  $-2,053$ )
- **HDD long-run scarring** ( $-4,982$  jobs at  $e = 10$ ;  $+4.9$ pp debt share at  $e = 10$ ) was not visible in the previous within-county LP framework
- Drought\_Exit shows **partial income scarring** — recovery rebounds but does not fully restore

## Resolved puzzle

- Cold-shock counties **recover on exit** (no within-county scarring)
- But the ever-treated population **ends up worse off** long-run (DiD)
- Both findings are correct; they measure different things

## Methodological closure

- Random effects rejected; FE is the right spec
- AQI exclusion from DiD now defensible (3.1% never-exposed)

# Open Caveats and Residual Risks

What the committee should know is still not addressed

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## Humidity confounding

- PRISM `tdmean` not yet a control
- Headline drought lag-2 and cold lag-1 effects **have not been stress-tested** against humidity
- Mitigation: Phase 4 resumption plan in place

## CDD CS-DiD region confound

- Treatment concentrated in southern states; controls in north
- Even two-way FE leaves regional secular trends
- Reported as suggestive;

## Multiple testing

- 427 ATT(`g,t`) cells  $\times$  7 outcomes  $\times$  up to 11 event-times
- Headline cells survive Bonferroni within-shock
- Smaller-cohort cells do not — flagged in synthesis

## County-master deduping

- Master is not strict one-row-per-(`fips`, `year`) due to rating-area joins
- Deduped in every committee-feedback script

# Thank you.

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## Discussion and questions

Track artifacts at

`conductor/tracks/committee_feedback_april_2026/`